

Pressure in gases

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

QUESTION		ANSWER	✓
1	State the effect on the volume of a gas of increasing its pressure at constant temperature.	The volume decreases.	☐
2	State the effect on the pressure of a gas of decreasing its volume at constant temperature.	The pressure increases.	☐
3	Explain why increasing the volume of a gas decreases its pressure.	The particles are more spread out, so they hit the walls less often.	☐
4	State the SI unit of pressure.	Pascal (Pa)	☐
5	State the SI unit of volume.	Cubic metre (m ³)	☐
6	State the equation for a fixed mass of gas at constant temperature.	pressure × volume = constant ($pV = \text{constant}$)	☐
7	State the direction of the net force that gas pressure exerts on a container wall.	At right angles to the wall.	☐
8	The volume of a gas is doubled at constant temperature. State what happens to its pressure.	It halves.	☐